LINEZOLID

MAOI activity — tyramine/serotonin

Weekly CBC

CHEMICAL NAME N-{[(5S)-3-[3-fluoro-4-(morpholin-4-yl)phenyl]-2-oxo-1,3-oxazolidin-5-yl]methyl}acetamide

CLASS Oxazolidinone antibiotic

TRADE NAMES Zyvox · Zyvoxam · Linospan

THERAPEUTIC USES / INDICATIONS

- MRSA pneumonia and complicated skin/soft-tissue infections.
- Vancomycin-resistant enterococcal (VRE) infections.
- MDR/XDR tuberculosis (specialist).
- Gram-positive infections when vancomycin unsuitable.

ADMINISTRATION

- **PO** or **IV** — 100% oral bioavailability (switch early to oral).
- IV infusion over 30–120 min.
- Oral suspension: invert gently (do not shake).

DOSAGES

- **Adult:** 600 mg every 12 h PO or IV for 10–14 days (max 28 days usually).
- **Children <12 yr:** 10 mg/kg every 8 h (max 600 mg/dose).
- **MDR-TB:** 600 mg once daily (specialist).
- No renal adjustment.

PHARMACOKINETICS

- **Peak:** PO 1–2 h. Bioavailability ~100%.
- Non-enzymatic oxidation (not CYP).
- Excreted in urine. **t½:** 5–7 h.

PHARMACODYNAMICS

- Binds **23S rRNA of 50S subunit** → prevents formation of 70S initiation complex → bacteriostatic (bactericidal against streptococci).
- Weak, reversible MAO inhibitor.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, headache.
- Taste disturbance, tongue discolouration.
- Candidiasis.

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (thrombocytopenia, anaemia); especially >2 weeks.
- Serotonin syndrome with SSRIs, SNRIs, tramadol, pethidine.
- Peripheral and optic neuropathy (long courses).
- Lactic acidosis; hypoglycaemia (in diabetics on insulin/oral agents).

CONTRAINDICATIONS & CAUTIONS

- Use with or within 2 weeks of MAOIs.
- Uncontrolled hypertension, phaeochromocytoma, thyrotoxicosis, carcinoid (unless monitored).
- **Caution:** pre-existing myelosuppression; serotonergic drugs.

KEY INTERACTIONS

- **SSRIs, SNRIs, TCAs, triptans, pethidine, tramadol:** serotonin syndrome.
- **Sympathomimetics (pseudoephedrine, adrenaline, dopamine):** hypertensive response.
- **Tyramine-rich foods:** hypertensive crisis (large amounts).

NURSING MANAGEMENT

- Review all medications for serotonergic/adrenergic drugs.
- Weekly CBC; monitor BP.
- Ask about visual changes and numbness in long courses.
- Monitor lactate if nausea/vomiting, acidosis signs.
- Monitor glucose in diabetics.

PATIENT EDUCATION

- Avoid large amounts of tyramine-rich foods (aged cheese, cured meat, soy sauce, draught beer).
- Avoid cold remedies with pseudoephedrine.
- Report blurred vision, numbness, bruising, agitation or fever.
- Oral tablets work as well as the drip.

KEY POINTS

- Oxazolidinone for MRSA and VRE.
- MAOI: serotonin syndrome, tyramine.
- 100% oral bioavailability.
- Myelosuppression and neuropathy in long courses.

SOURCES CHECKED **[183]** Linezolid 600 mg Film-Coated Tablets (UK SmPC) **[184]** Linezolid 2mg/ml Solution for Infusion (UK SmPC) **[185]** Zyvox, oral, IV (US label) **[186]** Linezolid, IV (US label)

Full references p. 432–457